NAME : Bakhtawar Munir

CLASS : BSCS-E

CMS-ID: 588984

ASSIGNMENT # 01

FUNDAMENTALS OF COMPUTER PROGRAMMING

COURSE INSTRUCTOR : DR.SHAH KHALID

Q1: Write a C++ program that accepts the following information about a student: Marks (25)

1) PROBLEM ANALYSIS:

Input:

Student registration number
Student name
Marks in programming fundamentals (out of 100)
Marks in mathematics (out of 100)
Marks in English (out of 100)

Process:

Add all 3 subject marks to find total marks
Divide total marks by 3 to find average marks
Calculate percentage using formula: (Total Marks / 300)*100

Output:

Display student info ,total marks, average marks, and percentage(formatted to 2 decimal places)

2) ALGORITHM :

Start

Ask user for student registration number and name

Ask user for marks in 3 subjects

Calculate total marks= pf marks + math marks + English marks

Calculate average marks= total marks / 3

Calculate percentage= (total marks / 300)*100

Show student info, total, average, and percentage with 2 decimal places

Stop

3) PSEUDOCODE :

START

INPUT reg num

INPUT student name

INPUT pf marks, math marks, Eng marks

total marks = pf marks + math marks + eng marks

average marks = total marks/3

percentage = (total marks / 300)*100

PRINT reg num, student name

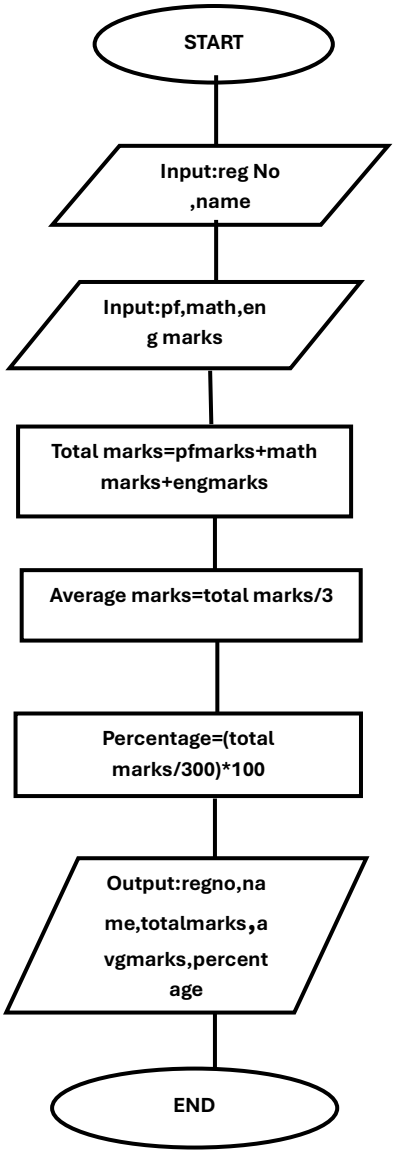
PRINT total marks

PRINT average marks (up to 2 decimals)

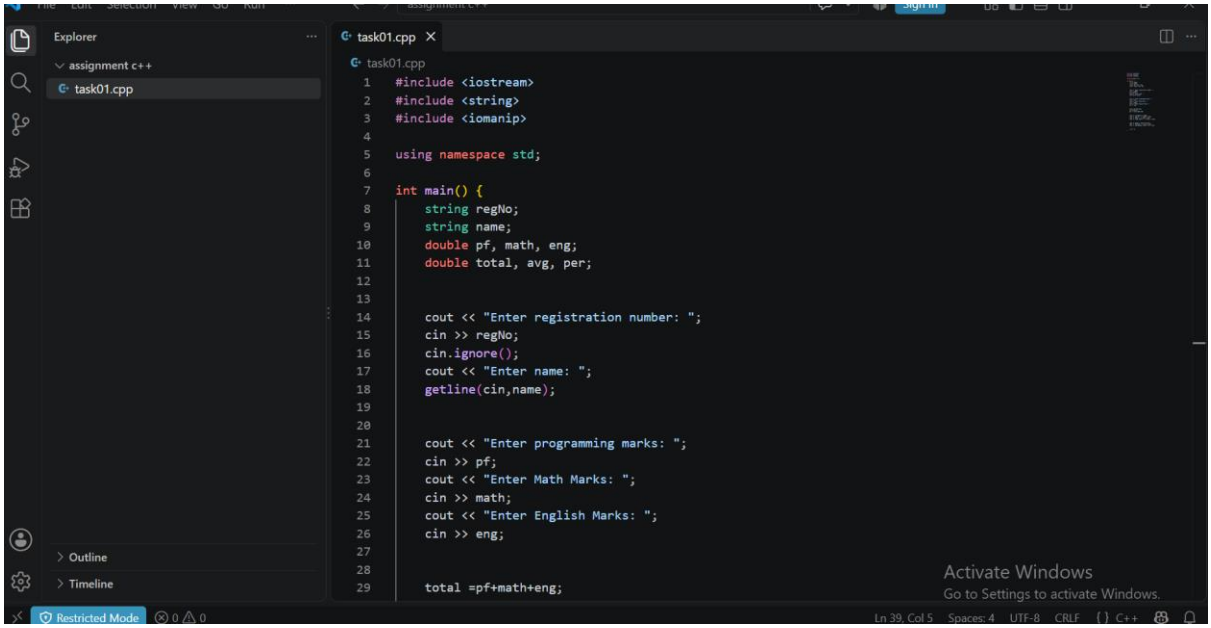
PRINT percentage (up to 2 decimals)

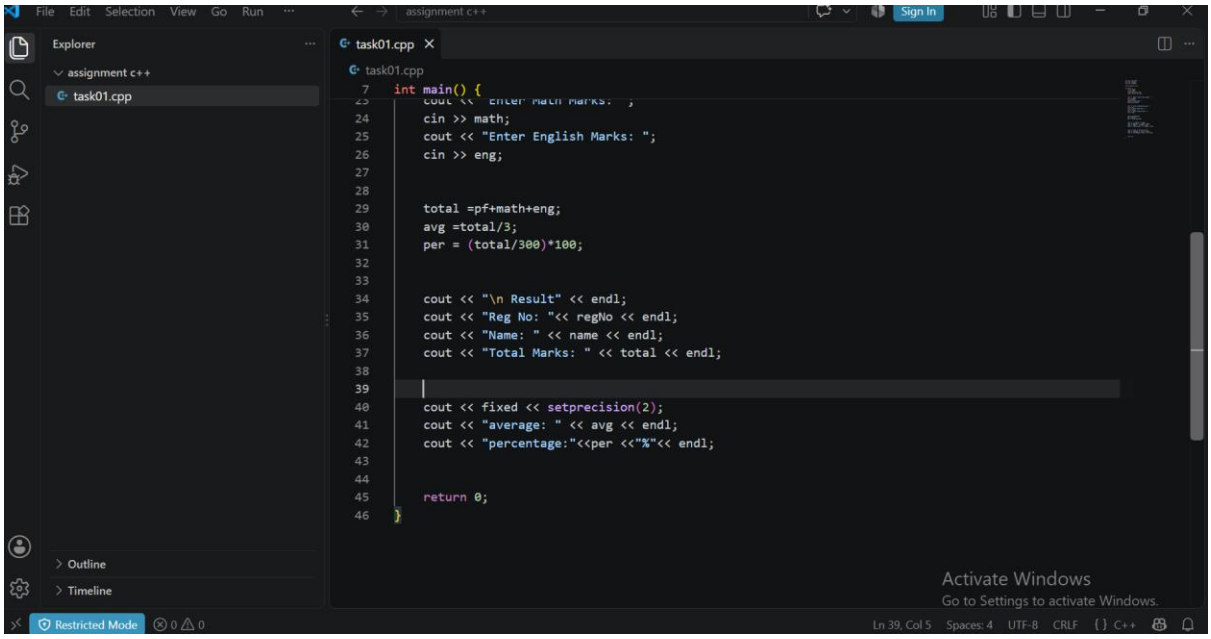
END

4) FLOW CHART :

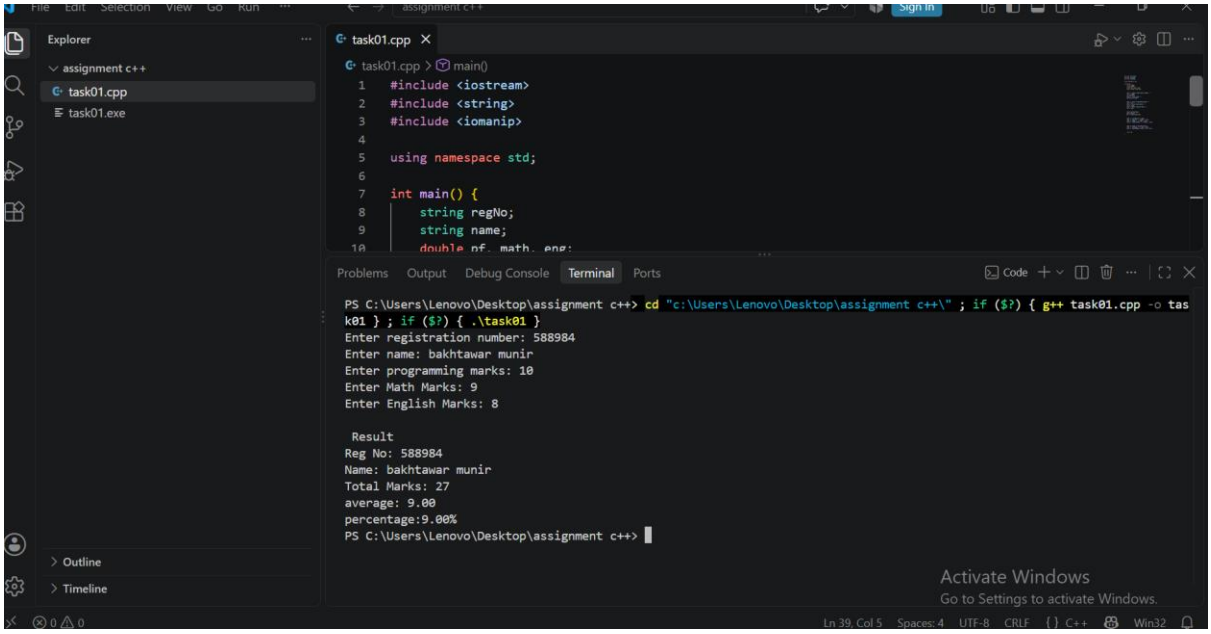


5) C++ SOURCE CODE :


Explorer
assignment c++
task01.cpp
Outline
Timeline
task01.cpp
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
int main() {
string regNo;
string name;
double pf, math, eng;
double total, avg, per;
cout << "Enter registration number: ";
cin >> regNo;
cin.ignore();
cout << "Enter name: ";
getline(cin,name);
cout << "Enter programming marks: ";
cin >> pf;
cout << "Enter Math Marks: ";
cin >> math;
cout << "Enter English Marks: ";
cin >> eng;
total =pf+math+eng;
Activate Windows
Go to Settings to activate Windows.
Restricted Mode
Ln 39, Col 5
Spaces: 4
UTF-8
CRLF
C++


File Edit Selection View Go Run
Sign In
Explorer
assignment c++
task01.cpp
Outline
Timeline
task01.cpp
int main() {
cin >> math;
cout << "Enter English Marks: ";
cin >> eng;
total =pf+math+eng;
avg =total/3;
per = (total/300)*100;
cout << "\n Result" << endl;
cout << "Reg No: "<< regNo << endl;
cout << "Name: " << name << endl;
cout << "Total Marks: " << total << endl;
cout << fixed << setprecision(2);
cout << "average: " << avg << endl;
cout << "percentage:"<<per <<"%"<< endl;
return 0;
}
Activate Windows
Go to Settings to activate Windows.
Restricted Mode
Ln 39, Col 5
Spaces: 4
UTF-8
CRLF
C++

6) SCREENSHOT OF OUTPUT :



Q2: Write a C++ program that accepts a duration in seconds and converts it into: Marks (25)

1) Problem Analysis :

Input:

Total duration in seconds

Process:

Divide total seconds by 3600 to get hours (total seconds/3600)

Take remainder (total seconds % 3600) to get remaining seconds

Divide remaining seconds by 60 to get Minutes (remaining seconds/60)

Take remainder (remaining seconds%60) to get final seconds

Divide total seconds by 60to get total minutes representation (total seconds/60)

Output:

Display hours,minutes,and seconds

Display total minutes

2) ALGORITHM :

Start

Get total seconds from user

Calculate hours=total seconds/3600

Calculate rem sec=total seconds%3600

Calculate minutes=rem sec/60

Calculate seconds = rem sec%60

Calculate total minutes=total seconds/60

Print hours,minutes,seconds and total minutes

Stop

3) PSEUDOCODE :

START

INPUT total seconds

hours=total seconds/3600

rem sec=total seconds%3600

minutes=rem sec/60

seconds=rem sec%60

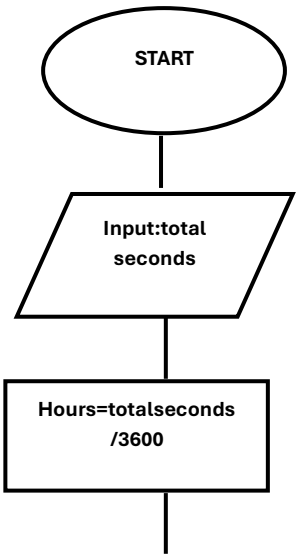
total minutes=total seconds/60

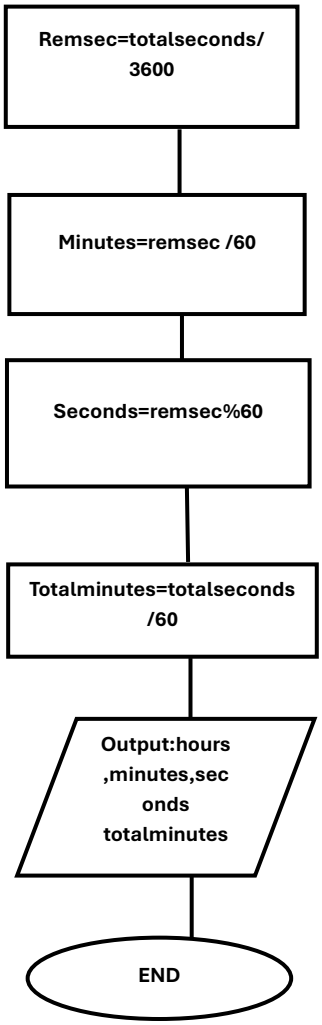
PRINT hours,minutes,seconds

PRINT total minutes

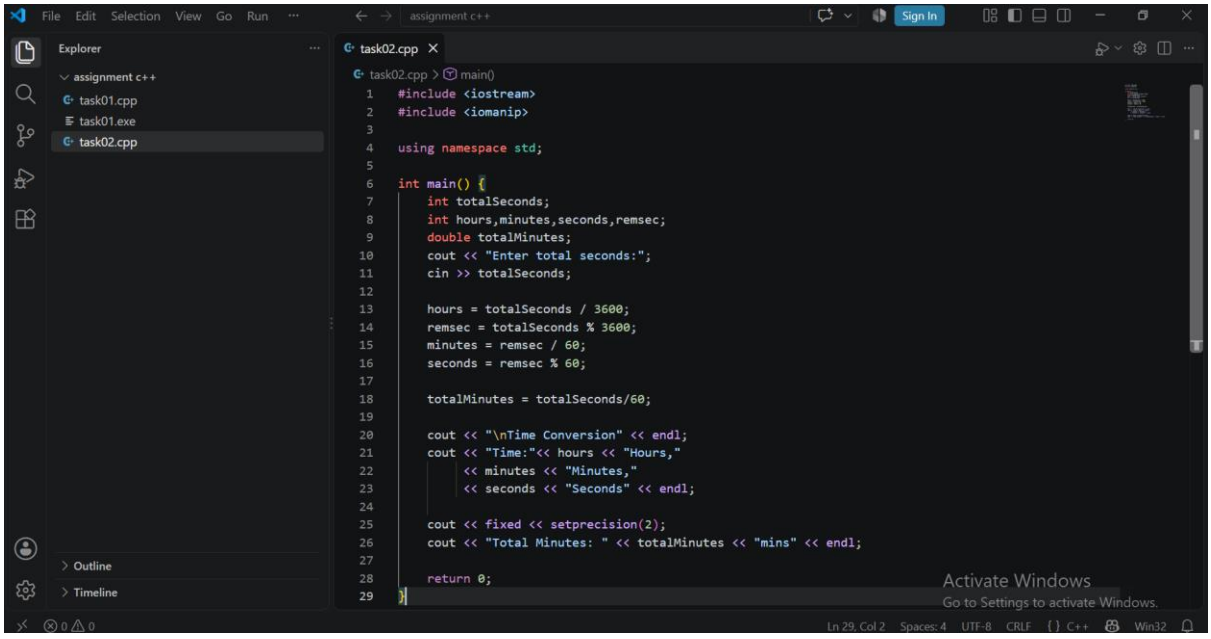
END

4) FLOWCHART :

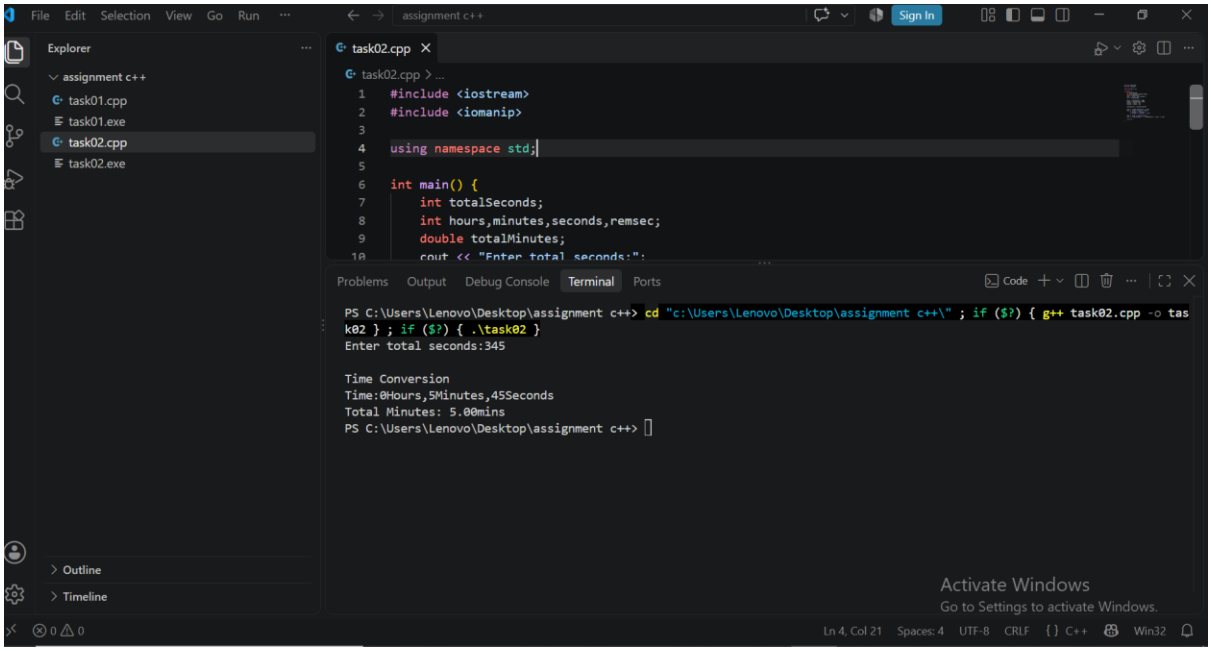




5) C++ SOURCE CODE :



6) SCREENSHOT OF OUTPUT :



Q3: Write a C++ program that generates a simple salary slip for an employee. Marks (25)

1) Problem Analysis :

Input:

Employee name

Employee id

Basic salary

Process:

House rent=basic salary*0.2

Medical allowance=basic salary*0.1

Transport allowance=basic salary*0.05

Gross salary=basic salary+house rent+medical allowance+transport allowance

tax=gross salary*0.05

net salary=gross salary-tax

Output:

Display properly formatted salary slip showing all allowances, gross salary, tax deduction, and net salary up to 2 decimal places

2) ALGORITHM :

Start

Get employee name,employee id, and basic salary from user.

Calculate house rent allowance (20% of basic salary)

Calculate medical allowance (10% of basic salary)

Calculate transport allowance (5% of basic salary)

Calculate gross salary by adding basic salary and all allowances

Calculate tax deduction (5% of gross salary)

Calculate net salary by subtracting tax from gross salary

Print salary slips with values rounded to 2 decimal places

Stop

3) Pseudocode :

START

INPUT emp name

INPUT emp id

INPUT basic salary

House rent=basic salary*0.2

medical=basic salary*0.1

transport=basic salary*0.05

gross salary=basic salary+house rent+medical+transport

tax=gross salary*0.05

net salary=gross salary-tax

PRINT emp name, emp id

PRINT basic salary

PRINT house rent, medical, transport

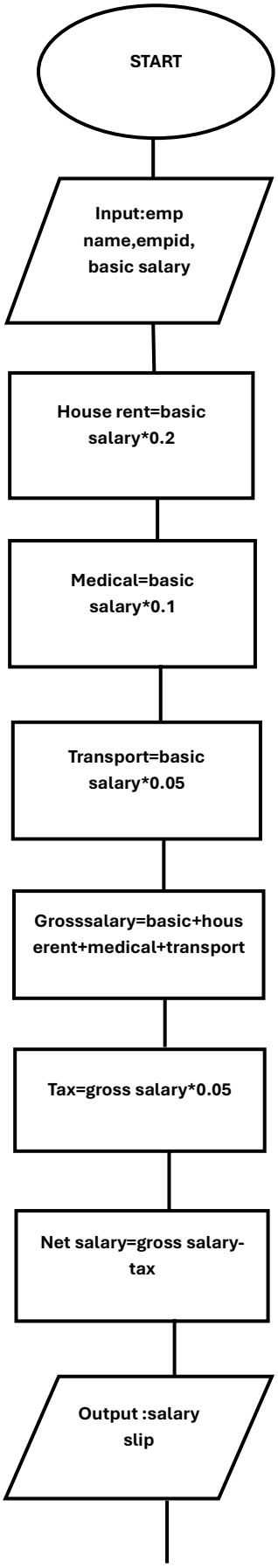
PRINT gross salary

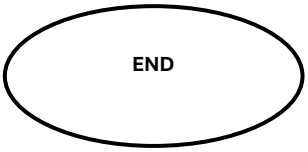
PRINT tax

PRINT net salary

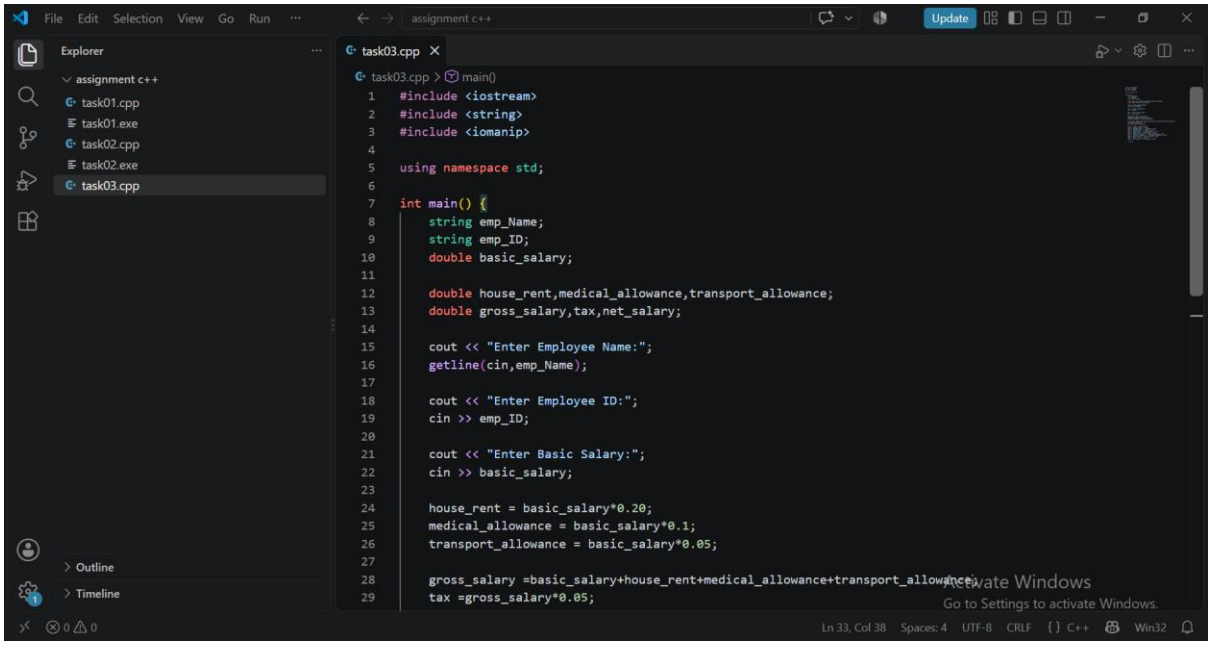
END

4) FLOW CHART :





5) C++ SOURCE CODE :



```cpp
#include <iostream>
#include <string>
#include <iomanip>

using namespace std;

int main() {
    string emp_Name;
    string emp_ID;
    double basic_salary;

    double house_rent,medical_allowance,transport_allowance;
    double gross_salary,tax,net_salary;

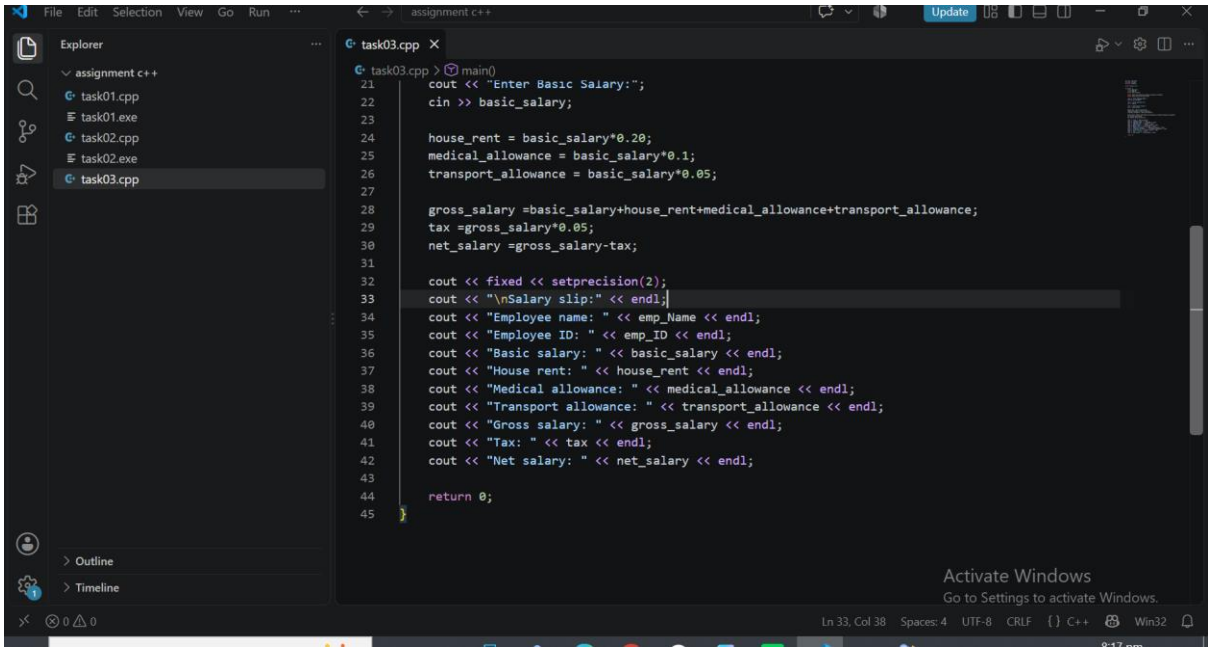
    cout << "Enter Employee Name:";
    getline(cin,emp_Name);

    cout << "Enter Employee ID:";
    cin >> emp_ID;

    cout << "Enter Basic Salary:";
    cin >> basic_salary;

    house_rent = basic_salary*0.20;
    medical_allowance = basic_salary*0.1;
    transport_allowance = basic_salary*0.05;

    gross_salary =basic_salary+house_rent+medical_allowance+transport_allowance;
    tax =gross_salary*0.05;
```



```cpp
    cout << "Enter Basic Salary:";
    cin >> basic_salary;

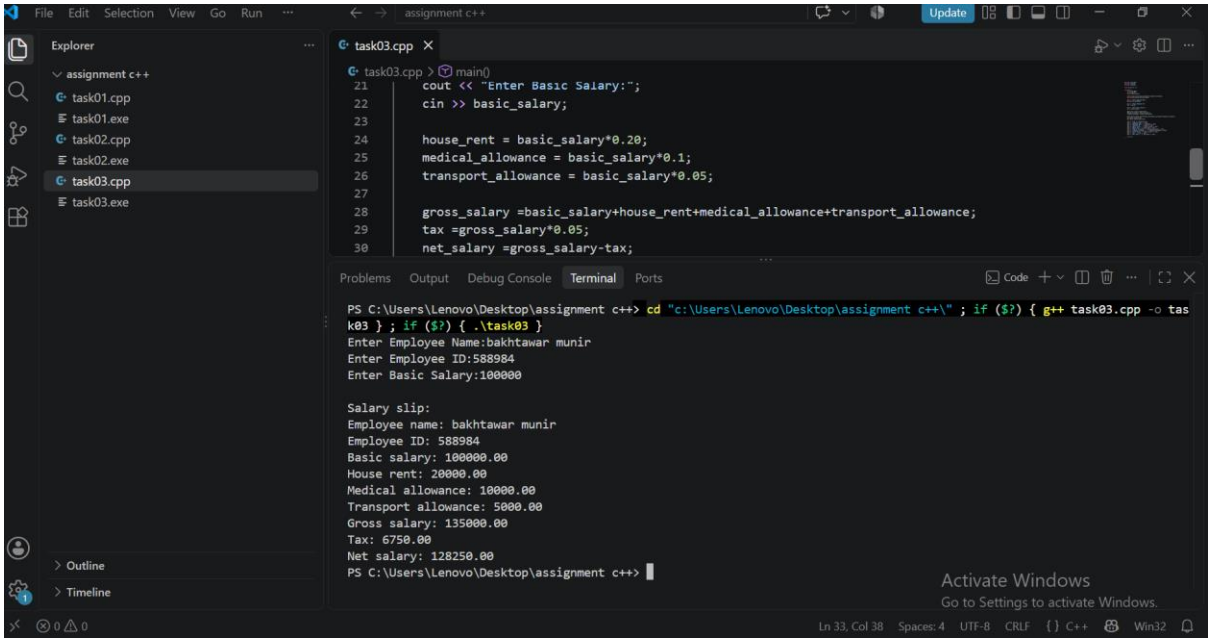
    house_rent = basic_salary*0.20;
    medical_allowance = basic_salary*0.1;
    transport_allowance = basic_salary*0.05;

    gross_salary =basic_salary+house_rent+medical_allowance+transport_allowance;
    tax =gross_salary*0.05;
    net_salary =gross_salary-tax;

    cout << fixed << setprecision(2);
    cout << "\nSalary slip:" << endl;
    cout << "Employee name: " << emp_Name << endl;
    cout << "Employee ID: " << emp_ID << endl;
    cout << "Basic salary: " << basic_salary << endl;
    cout << "House rent: " << house_rent << endl;
    cout << "Medical allowance: " << medical_allowance << endl;
    cout << "Transport allowance: " << transport_allowance << endl;
    cout << "Gross salary: " << gross_salary << endl;
    cout << "Tax: " << tax << endl;
    cout << "Net salary: " << net_salary << endl;

    return 0;
}
```

6) SCREEN SHOT OF OUTPUT :



Q4: A bank wants to generate a summary of a customer's account transaction. Marks (25)

1) Problem Analysis :

Input:

Customer name

Account number

Opening balance

Deposit amount

Withdrawal amount

Interest Rate %

Process:

Balance after tx=opening balance+deposit-withdrawal

interest=balance after tx*(interest rate/100)

final balance=balance after tx+interest

Currency denomination calculations on int amount =final balance

Notes 5000=int amount/5000

rem=int amount%5000

notes 100=rem/100

remaining amount=(final balance-(int amount-(rem%100)))

total tx amount=deposit+withdrawal

deposit percent=(deposit/opening balance)*100

withdrawal percent=(withdrawal / opening balance)*100

Output:

Display all details, calculated balances,interest,currency notes count,remaining amount and percentages (round to 2 dp)

2) ALGORITHM :

Start

Get customer name,account number,opening balance,deposit,withdrawal and interest rate

Calculate balance after transactions(Opening+Deposit-Withdrawal).

Calculate interest amount (balance*Rate/100)

Calculate final balance (balance+Interest)

Calculate rs.5000,1000,500,and 100 currency notes using integer division and modulus

Calculate the remaining amount after currency distribution

Calculate total transaction amount (Deposit+Withdrawal)

Calculate deposit and withdrawal percentages relative to opening balance

Display the full bank transaction summary

Stop

3) PSEUDOCODE :

START

INPUT name, acc num

INPUT opening Bal,deposit,withdrawal,rate

Bal after tx=opening Bal+deposit-withdrawal

interest = Bal after tx*(rate / 100)

final Bal=bal after tx + interest

int Bal=INT(final bal)

n5000=int Bal/5000

rem=int Bal%5000

n1000=rem/1000

rem=rem%1000

n500=rem/500

rem=rem%500

n100=rem/100

rem Bal=final Bal-(n5000*5000+n1000*1000 +n500*500 + n100*100)

total tx=deposit+withdrawal

dep percent =(deposit /opening bal)*100

withPercent = (withdrawal / openingBal)*100

PRINT name, acc num

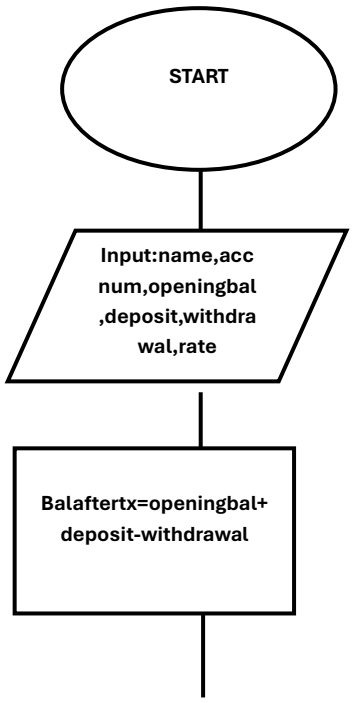
PRINT bal after tx,interest,final Bal

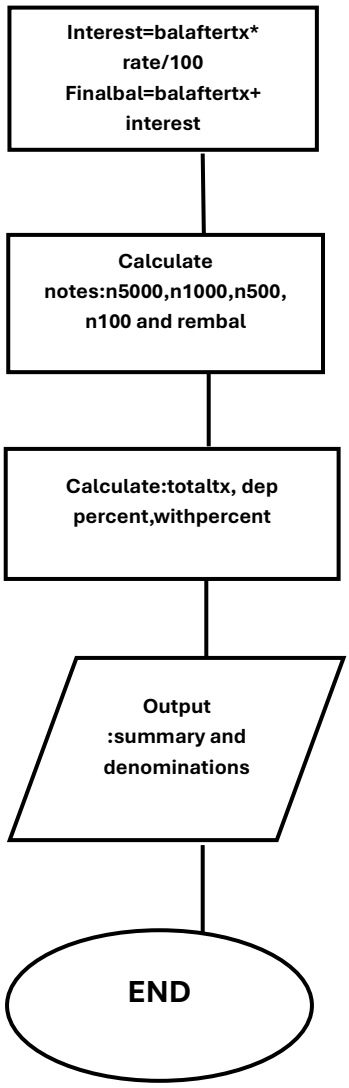
PRINT n5000, n1000, n500, n100, remBal

PRINT total tx,dep percent,with percent

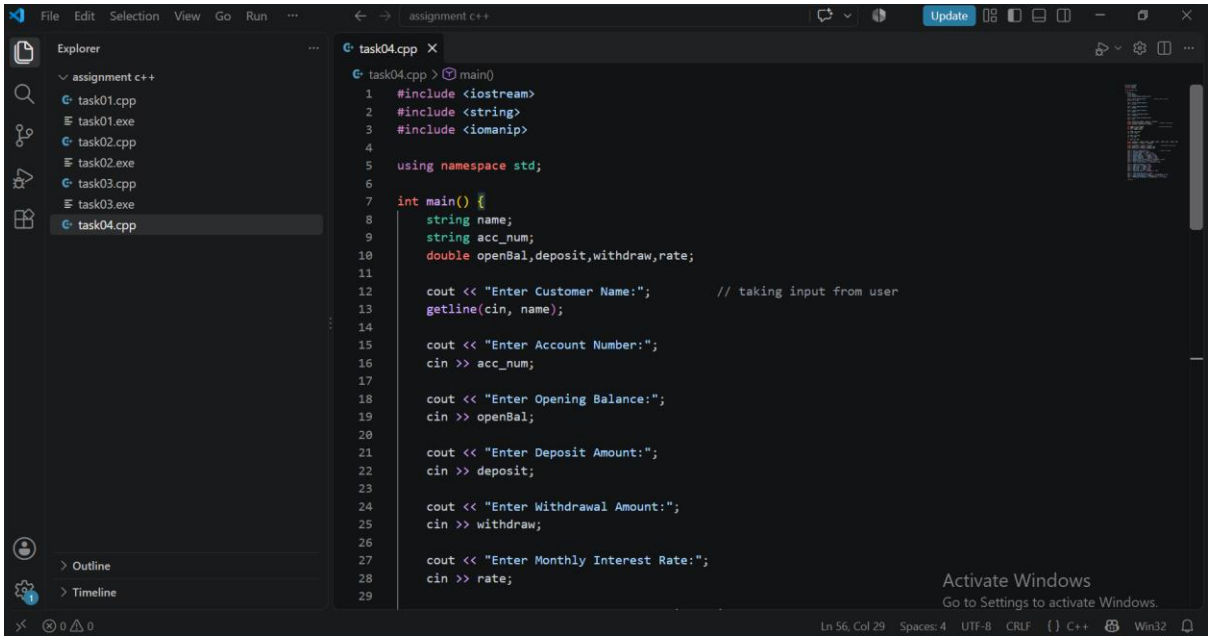
END

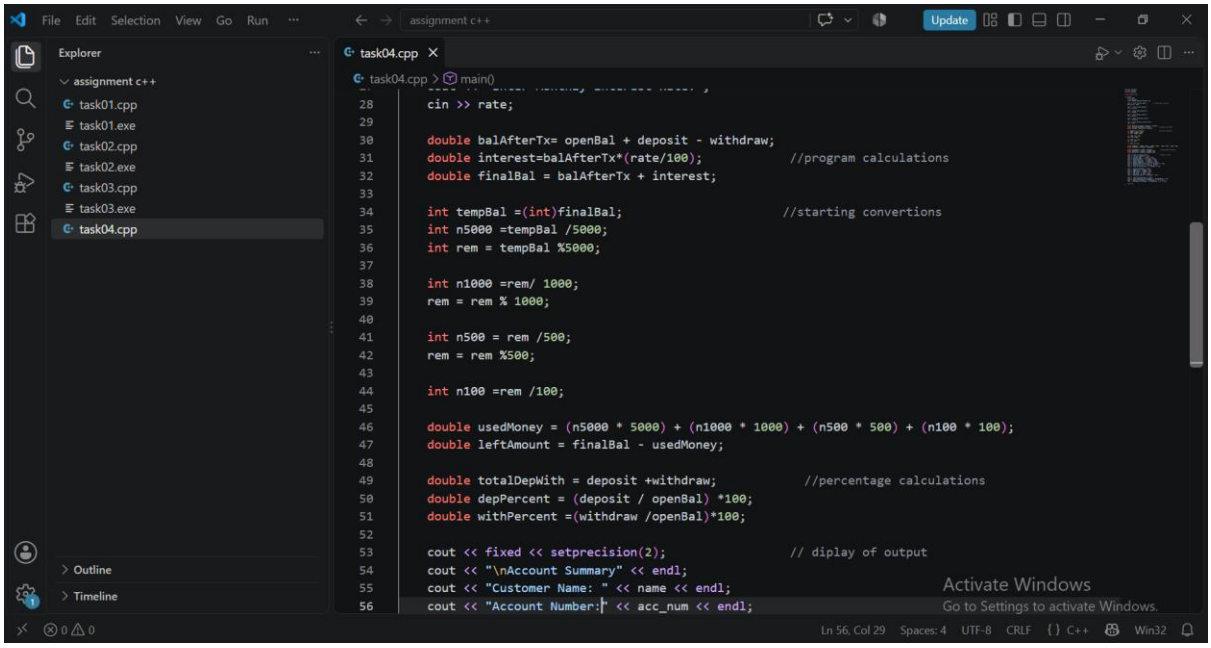
4) FLOW CHART :





5) C++ SOURCE CODE :





```
    cin >> rate;

    double balAfterTx= openBal + deposit - withdraw;
    double interest=balAfterTx*(rate/100);          //program calculations
    double finalBal = balAfterTx + interest;

    int tempBal =(int)finalBal;                     //starting convertions
    int n5000 =tempBal /5000;
    int rem = tempBal %5000;

    int n1000 =rem/ 1000;
    rem = rem % 1000;

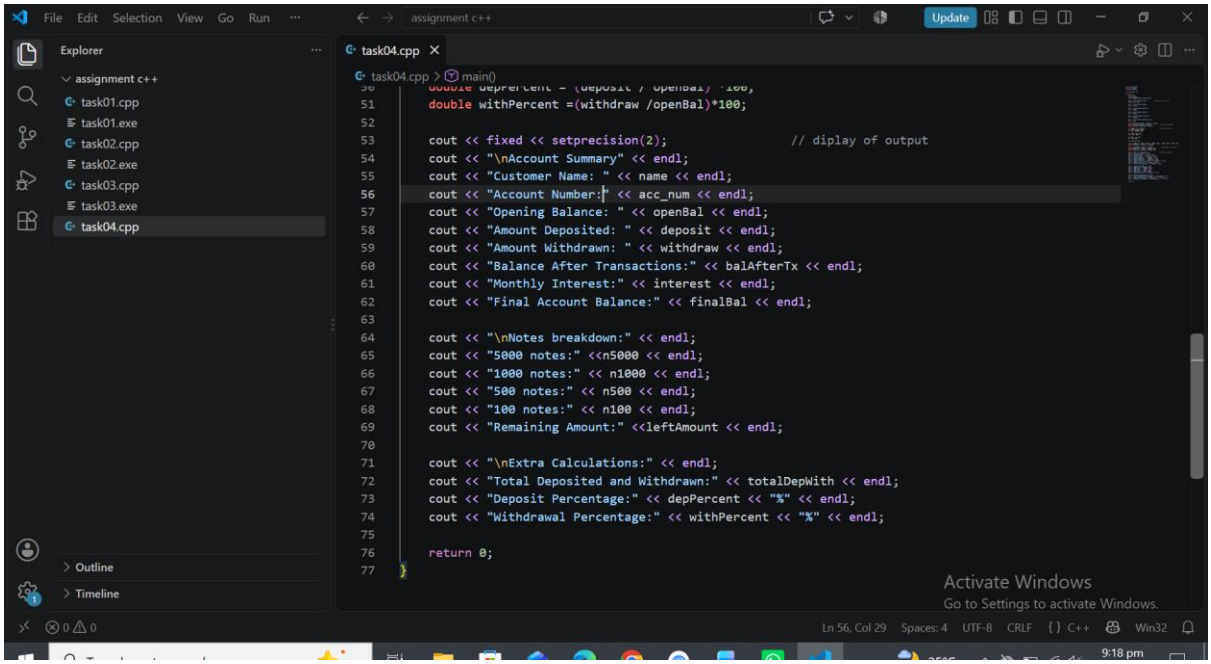
    int n500 = rem /500;
    rem = rem %500;

    int n100 =rem /100;

    double usedMoney = (n5000 * 5000) + (n1000 * 1000) + (n500 * 500) + (n100 * 100);
    double leftAmount = finalBal - usedMoney;

    double totalDepWith = deposit +withdraw;            //percentage calculations
    double depPercent = (deposit / openBal) *100;
    double withPercent =(withdraw /openBal)*100;

    cout << fixed << setprecision(2);               // diplay of output
    cout << "\nAccount Summary" << endl;
    cout << "Customer Name: " << name << endl;
    cout << "Account Number:" << acc_num << endl;
```



```
    double withPercent =(withdraw /openBal)*100;

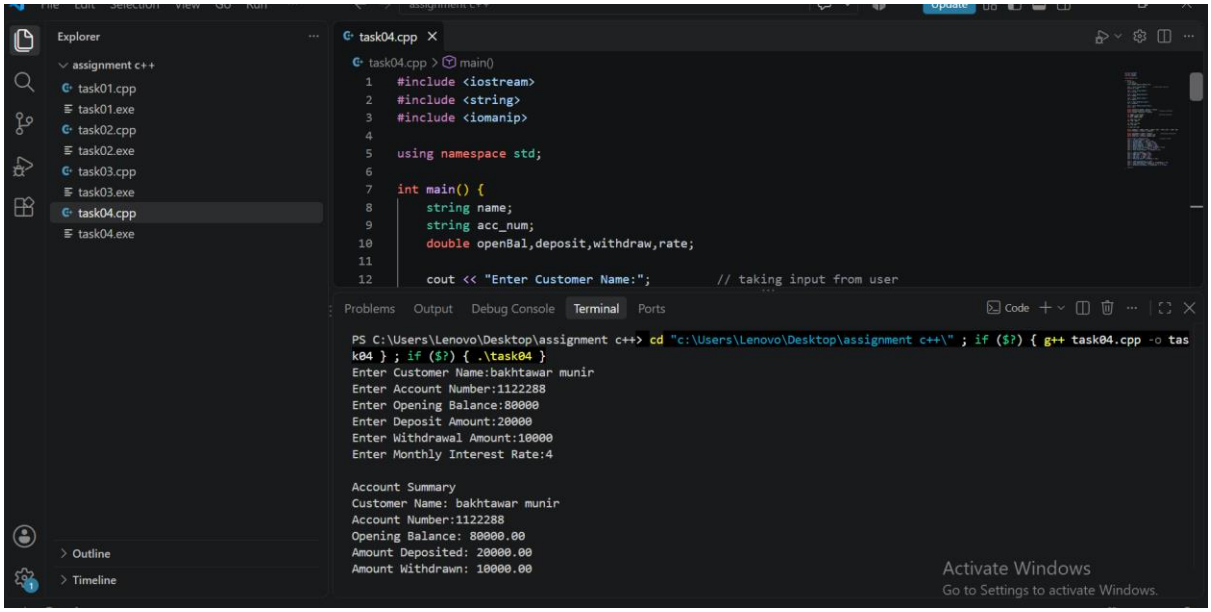
    cout << fixed << setprecision(2);               // diplay of output
    cout << "\nAccount Summary" << endl;
    cout << "Customer Name: " << name << endl;
    cout << "Account Number:" << acc_num << endl;
    cout << "Opening Balance: " << openBal << endl;
    cout << "Amount Deposited: " << deposit << endl;
    cout << "Amount Withdrawn: " << withdraw << endl;
    cout << "Balance After Transactions:" << balAfterTx << endl;
    cout << "Monthly Interest:" << interest << endl;
    cout << "Final Account Balance:" << finalBal << endl;

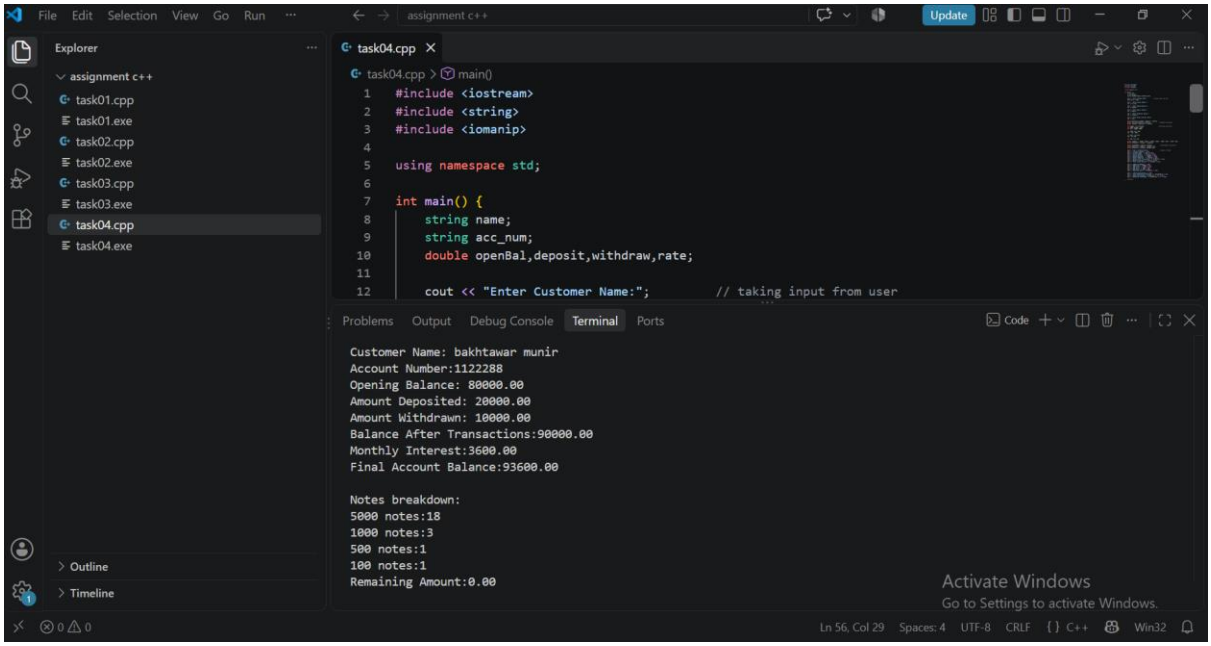
    cout << "\nNotes breakdown:" << endl;
    cout << "5000 notes:" <<n5000 << endl;
    cout << "1000 notes:" << n1000 << endl;
    cout << "500 notes:" << n500 << endl;
    cout << "100 notes:" << n100 << endl;
    cout << "Remaining Amount:" <<leftAmount << endl;

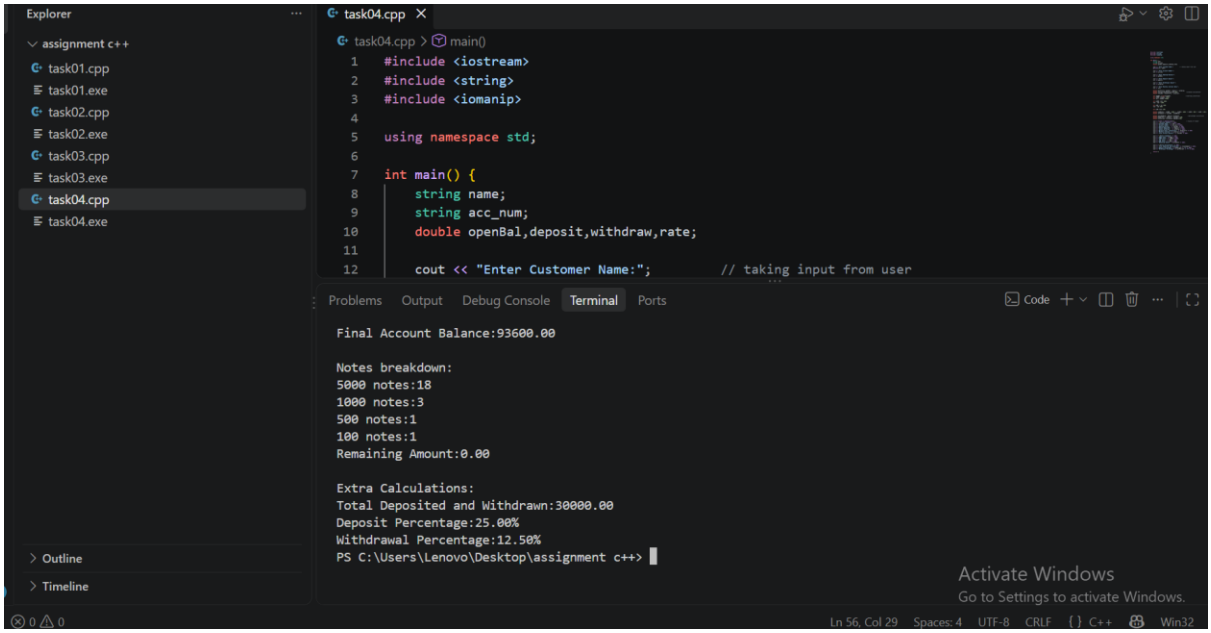
    cout << "\nExtra Calculations:" << endl;
    cout << "Total Deposited and Withdrawn:" << totalDepWith << endl;
    cout << "Deposit Percentage:" << depPercent << "%" << endl;
    cout << "Withdrawal Percentage:" << withPercent << "%" << endl;

    return 0;
}
```

6) SCREEN SHOT OF OUTPUT :


Explorer
assignment c++
task01.cpp
task01.exe
task02.cpp
task02.exe
task03.cpp
task03.exe
task04.cpp
task04.exe
Outline
Timeline
task04.cpp
task04.cpp > main()
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
int main() {
string name;
string acc_num;
double openBal,deposit,withdraw,rate;
cout << "Enter Customer Name:"; // taking input from user
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\Lenovo\Desktop\assignment c++> cd "c:\Users\Lenovo\Desktop\assignment c++\" ; if ($?) { g++ task04.cpp -o tas
k04 } ; if ($?) { .\task04 }
Enter Customer Name:bakhtawar munir
Enter Account Number:1122288
Enter Opening Balance:80000
Enter Deposit Amount:20000
Enter Withdrawal Amount:10000
Enter Monthly Interest Rate:4
Account Summary
Customer Name: bakhtawar munir
Account Number:1122288
Opening Balance: 80000.00
Amount Deposited: 20000.00
Amount Withdrawn: 10000.00
Activate Windows
Go to Settings to activate Windows.


File Edit Selection View Go Run
assignment c++
Update
Explorer
assignment c++
task01.cpp
task01.exe
task02.cpp
task02.exe
task03.cpp
task03.exe
task04.cpp
task04.exe
Outline
Timeline
task04.cpp
task04.cpp > main()
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
int main() {
string name;
string acc_num;
double openBal,deposit,withdraw,rate;
cout << "Enter Customer Name:"; // taking input from user
Problems Output Debug Console Terminal Ports
Code
Customer Name: bakhtawar munir
Account Number:1122288
Opening Balance: 80000.00
Amount Deposited: 20000.00
Amount Withdrawn: 10000.00
Balance After Transactions:90000.00
Monthly Interest:3600.00
Final Account Balance:93600.00
Notes breakdown:
5000 notes:18
1000 notes:3
500 notes:1
100 notes:1
Remaining Amount:0.00
Activate Windows
Go to Settings to activate Windows.
Ln 56, Col 29 Spaces: 4 UTF-8 CRLF { } C++ Win32


Explorer
assignment c++
task01.cpp
task01.exe
task02.cpp
task02.exe
task03.cpp
task03.exe
task04.cpp
task04.exe
Outline
Timeline
task04.cpp
task04.cpp > main()
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
int main() {
string name;
string acc_num;
double openBal,deposit,withdraw,rate;
cout << "Enter Customer Name:"; // taking input from user
Problems
Output
Debug Console
Terminal
Ports
Code
Final Account Balance:93600.00
Notes breakdown:
5000 notes:18
1000 notes:3
500 notes:1
100 notes:1
Remaining Amount:0.00
Extra Calculations:
Total Deposited and Withdrawn:30000.00
Deposit Percentage:25.00%
Withdrawal Percentage:12.50%
PS C:\Users\Lenovo\Desktop\assignment c++>
Activate Windows
Go to Settings to activate Windows.
Ln 56, Col 29
Spaces: 4
UTF-8
CRLF
C++
Win32